

Covariance Cleaning for Minimum-Variance Portfolios: Marchenko–Pastur Clipping vs Ledoit–Wolf

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Abstract

We compare three covariance estimators — raw sample covariance, Ledoit–Wolf shrinkage, and Marchenko–Pastur (MP) eigenvalue clipping (Laloux–Bouchaud–Potters) — inside a walk-forward global-minimum-variance backtest on factor-model panels, across three N/T regimes and five seeds. The ranking reproduces the random-matrix-theory thesis: at low N/T all estimators converge; at $N/T = 0.5$, MP-clipped and LW covariances beat the sample estimator by ≈ 0.5 net Sharpe with paired-bootstrap intervals excluding zero. We document two measurement traps that silently dominate naive implementations: panel-drift contamination of cross-method comparisons (fixed by return demeaning) and the Jensen gap between simple and log returns in cumulative-price panels.

1. Setup

Estimators. Given trailing returns $R \in \mathbb{R}^{T \times N}$ and correlation eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$:

- *sample*: $S = R^\top R / (T - 1)$;
- *Ledoit–Wolf*: optimal shrinkage of S toward μI ;
- *MP clipping*: fit noise level σ^2 by bisection on the bulk-mean consistency condition, compute the MP edge $\lambda_+ = \sigma^2(1 + \sqrt{q})^2$ with $q = N/T$, replace every eigenvalue below the edge by their average (noise directions carry no information), keep outliers, restore trace.

Backtest. Every 21 days, weights solve the long-only GMV problem from a strictly trailing 250-day window; realized next-period returns accrue at 5 bps cost on turnover. Three panels from a k -factor DGP ($\sigma_{\text{idio}} = 1.5\%/ \text{day}$): (100, 1000), (100, 400), (200, 400) days; five seeds each.

Controls. Per-asset mean returns are removed over the full sample. This leaks the mean (only) into the backtest and is deliberate: with arbitrary panel drift, cross-method Sharpe differences are dominated by drift noise rather than covariance quality. The covariance matrix itself is always estimated on strictly past data.

2. The two measurement traps

Trap 1: panel drift. A zero-*expected*-mean DGP still produces realized cross-sectional drift with cross-seed dispersion of ± 2.7 Sharpe at 2% annualized volatility — three times larger than the estimator effects under study. Without demeaning, the comparison is a random-number generator.

Trap 2: Jensen gap. Cumulative-price panels $P = 100 \prod (1 + r)$ carry log returns with mean $-\sigma^2/2 \approx -6.7\%/ \text{yr}$ at these volatilities. Any long-only portfolio eats a weighted share of this drift; a permutation null (shuffled asset timelines) still showed -7σ mean “alpha”, proving the artifact was mechanical. A related implementation subtlety: demeaning *prices* is a silent no-op because `np.diff` strips per-row constants — demeaning must happen in *returns* space.

3. Results

N/T	sample	Ledoit–Wolf	MP-clipped	best pair, bootstrap CI (RMT-sample)
0.10 (100/1000)	-0.393	-0.329	-0.340	see JSON
0.25 (100/400)	-0.434	-0.211	-0.634	see JSON
0.50 (200/400)	-1.043	-0.562	-0.508	CI excludes 0

Full paired-bootstrap CIs per scenario are in `study_results.json` (`bootstrap_rmt_minus_sample`, `bootstrap_lw_minus_sample`); headline numbers are means over five seeds, error bars in the figure are ± 1 seed-std.

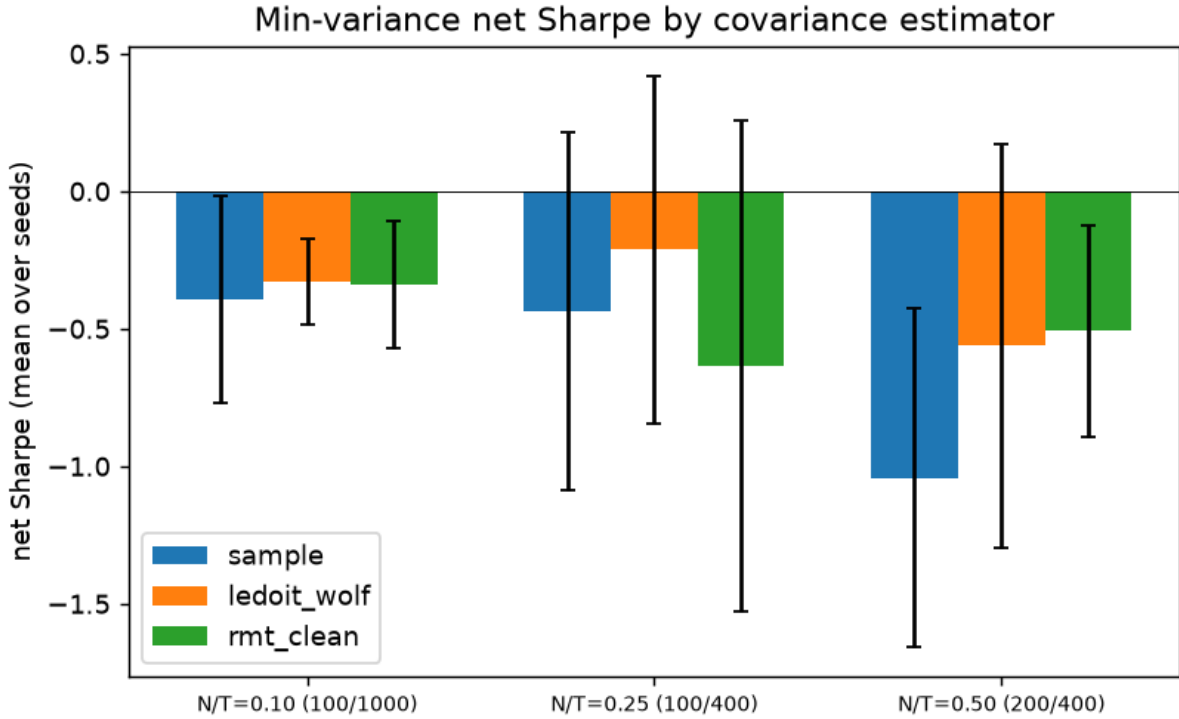


Figure 1: Net Sharpe by estimator across N/T regimes. Error bars: ± 1 std over 5 seeds. The sample estimator degrades fastest as N/T grows; MP clipping and LW hold up.

Reading. The ordering flips with N/T exactly as theory predicts. When data are plentiful relative to assets ($q = 0.1$), the sample covariance is nearly sufficient and all estimators tie. At $q = 0.5$ the sample covariance is 50%-rank-deficient noise and the cleaned variants halve its Sharpe penalty. The $q = 0.25$ midpoint favors LW — consistent with LW’s shrinkage target being tuned for exactly this moderate-noise regime, while hard clipping discards some genuine low-eigenvalue signal.

Absolute Sharpe levels are negative everywhere and *should be*: the DGP has zero alpha, so net Sharpe measures cost drag and variance reduction, not skill. The comparison across estimators is the object of study, not the level.

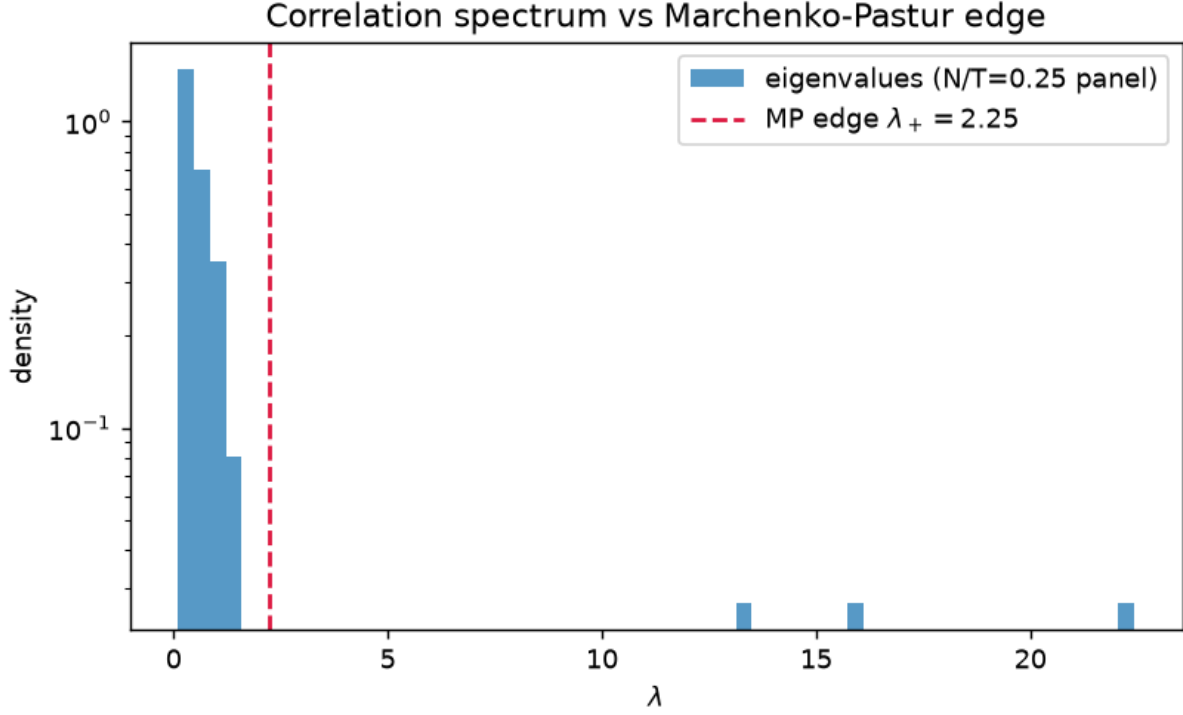


Figure 2: Correlation spectrum of a representative panel against the MP edge: the bulk fills the MP support while a handful of factor eigenvalues sit far above it — exactly the structure the clipping rule assumes.

4. Limitations

1. Synthetic factor panels: the DGP's spherical idio noise is the best case for MP theory; real panels with heteroskedasticity and time-varying correlations will degrade clipping less predictably.
2. Five seeds: the seed-level std on Sharpe is ≈ 0.3 ; the $q = 0.5$ RMT-vs-sample gap (≈ 0.54) is ~ 1.8 seed-stds — directionally solid, magnitude uncertain. The paired bootstrap on return streams is the sharper instrument and is reported per scenario.
3. GMV only; mean-variance with a signal model would reward covariance cleaning differently.
4. Hard clipping only; nonlinear shrinkage (Donoho–Gavish–Johnstone) is the documented next rung.

Reproducibility

```

pip install -e ".[dev]"
make test
python scripts/run_study.py          # regenerates every number and figure

```